

Find the driving-point impedance seen by the voltage source in Figure P11-2 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

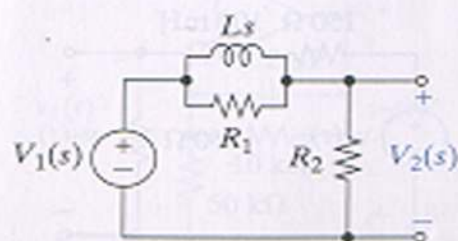


FIGURE P11-2

Find the driving-point impedance seen by the voltage source in Figure P11-4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

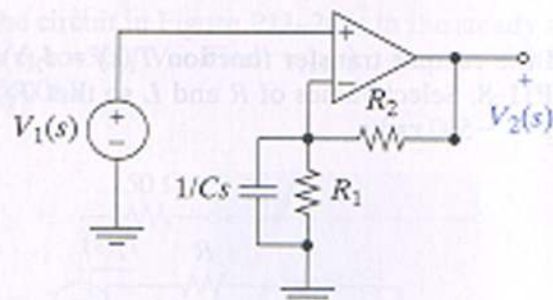


FIGURE P11-4

Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-7. Select values of R and C so that $T_V(s)$ has a pole at $s = -500$ rad/s.

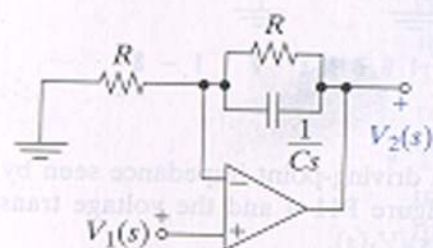


FIGURE P11-7

The circuit in Figure P11-28 is in the steady state with $i_1(t) = 10 \cos 500t$ mA, $R_1 = 100 \Omega$, $R_2 = 400 \Omega$, and $L = 100$ mH. Find $i_{2ss}(t)$. Repeat with $i_1(t) = 10 \cos 5000t$ mA.

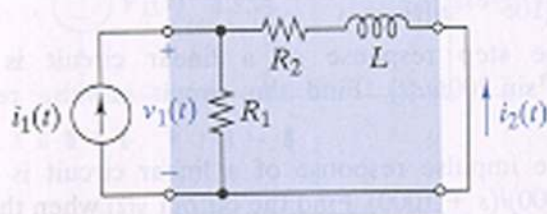


FIGURE P11-28

Design a circuit to realize the transfer function below using only resistors, capacitors, and not more than one OP AMP. Scale the circuit so that the final design uses only 20-k Ω resistors.

$$T_V(s) = \pm \frac{1000s}{(s+500)(s+1000)}$$

It is claimed that both circuits in Figure P11-58 realize the transfer function

$$T_V(s) = K \left(\frac{s+2000}{s+1000} \right)$$

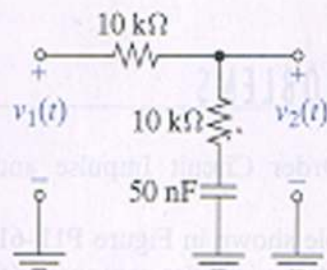
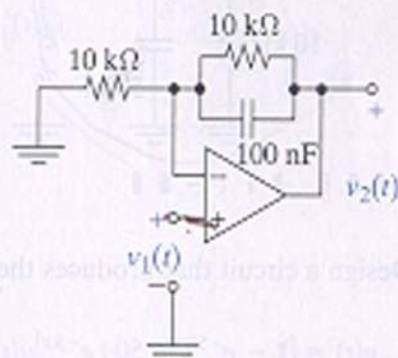


FIGURE P11-58

- Verify that both circuits realize the specified $T_V(s)$.
- Which circuit would you choose if the output must drive a 1-k Ω load?
- Which circuit would you choose if the input comes from a 50- Ω source?
- It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage. Do you agree or disagree? Explain.

12-4 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-4.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Draw the straight-line approximations of the gain and phase of $T_V(j\omega)$.
- What element values would you change to double the passband gain without changing the cutoff frequency?

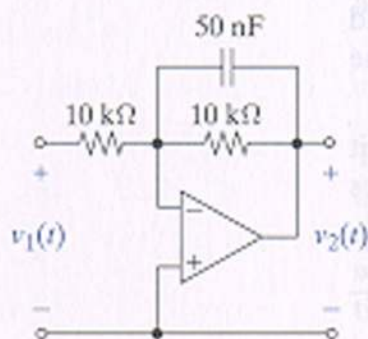


FIGURE P12-4

12-28 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ for the bandpass circuit in Figure P12-28. Find the parameters B , Q , ω_{C1} , and ω_{C2} for $R = 50 \Omega$, $L = 50 \mu\text{H}$, and $C = 20 \text{ nF}$.

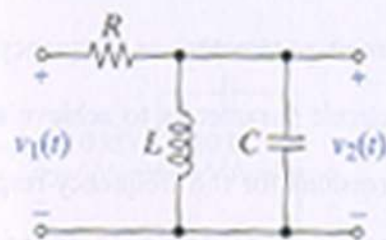


FIGURE P12-28

12-10 The transfer function of a first-order circuit is

$$T(s) = \frac{10/s}{100/s + 1/20}$$

- Identify the type of gain response. Find the cutoff frequency and the passband gain.
- Sketch the straight-line gain response. Use the straight-line gain to estimate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.

12-15 The circuit in Figure P12-15 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 10 and the cutoff frequencies are 100 rad/s and 2500 rad/s. Use practical element values with $R \geq 10 \text{ k}\Omega$ and $C \leq 1 \mu\text{F}$.

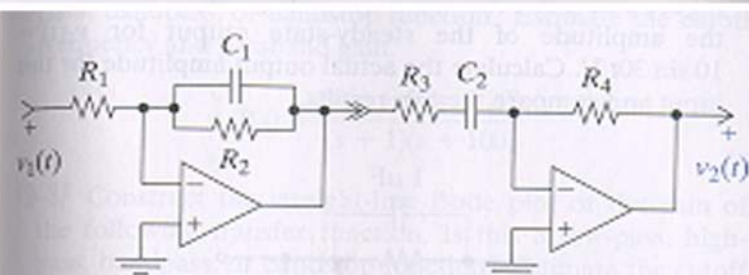


FIGURE P12-15